Digital Signal Processing

Lecture 6

Frequency Response of LTI System

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Partial Fraction Expansion in MATLAB

$$X(z)=\frac{1+3z^{-1}+3z^{-2}+z^{-3}}{1-3z^{-1}+2z^{-2}}=2.25+0.5z^{-1}+\frac{-8}{1-z^{-1}}+\frac{6.75}{1-2z^{-1}}$$

```
» [r,p,k]=residuez([1,3,3,1],[1,-3,2])
r =
   6.75000000000000
  -8.00000000000000
p =
     2
     1
k =
   2.25000000000000   0.50000000000000
```

MATLAB's `residuez` *can also go back the other way, and can also handle repeated roots*

– *Note: '*`residuez`*' is part of the optional Signal Processing Toolbox*

Selected z-Transform Theorems

- The delay or shift property:

$$x[n-n_0] \quad \Leftrightarrow \quad z^{-n_0} X(z)$$

- The convolution property:

$$y[n] = x[n] * h[n] \quad \Leftrightarrow \quad Y(z) = H(z)X(z)$$

- A consequence:

$$y[n] = x[n] * \delta[n-n_0] \quad \Leftrightarrow \quad Y(z) = z^{-n_0} X(z)$$

An IIR System

- Difference equation:

$$y[n] = ay[n-1] + x[n] \Leftrightarrow Y(z) = az^{-1}Y(z) + X(z)$$

- System function:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1}{1-az^{-1}} = \frac{z}{z-a} \quad |z|>|a|$$

ROC if causal

- Frequency response:

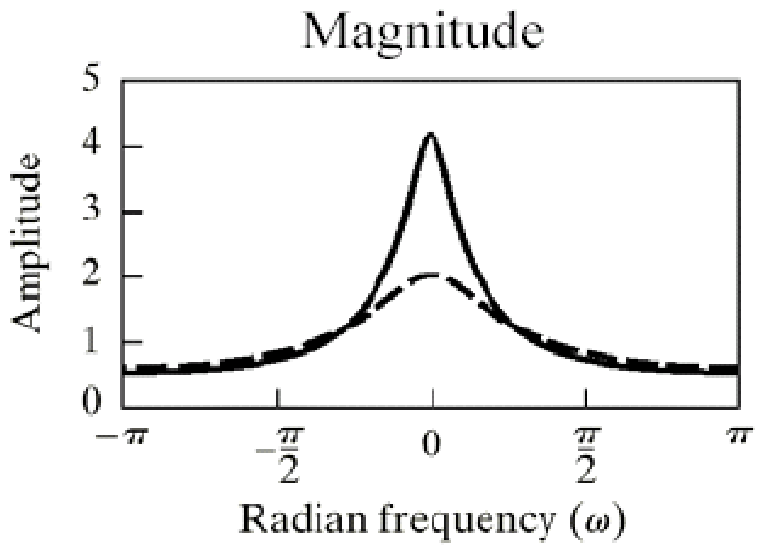
$$H(e^{j\omega}) = H(z)\big|_{z=e^{j\omega}} = \frac{1}{1-ae^{-j\omega}}$$

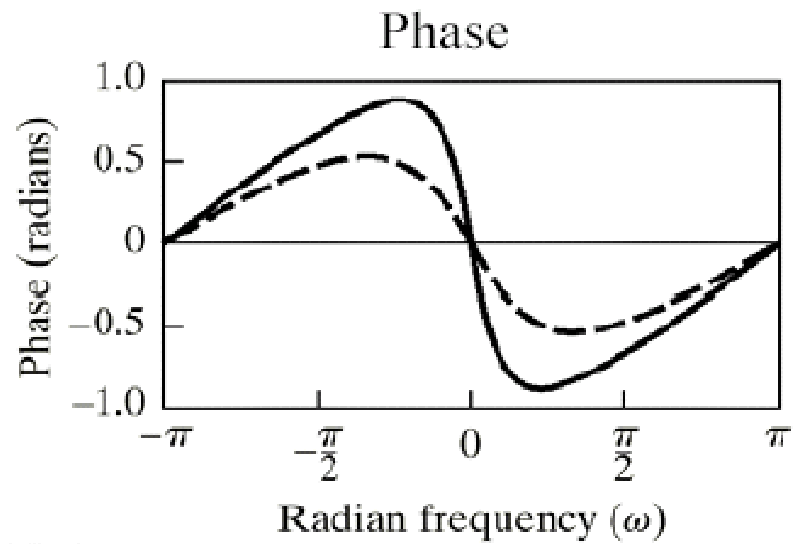
Assumes $|a|<1$

- Impulse response:

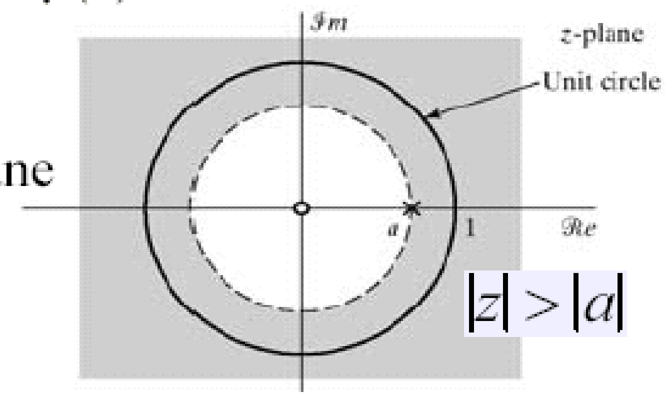
$$h[n] = a^n u[n]$$

IIR Frequency Response





z-plane



System Function Of a Difference Equation

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k]$$

$$\sum_{k=0}^{N} a_k Y(z) z^{-k} = \sum_{k=0}^{M} b_k X(z) z^{-k}$$

$$\left(\sum_{k=0}^{N} a_k z^{-k}\right) Y(z) = \left(\sum_{k=0}^{M} b_k z^{-k}\right) X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\left(\sum_{k=0}^{M} b_k z^{-k}\right)}{\left(\sum_{k=0}^{N} a_k z^{-k}\right)}$$

Difference equations give rise to rational z transforms!

H[z] and h[n]

- Consider a causal system; *i.e.* $h[n]=0$ for $n<0$:

$$H(z)=\frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}=\frac{b_0}{a_0}\frac{\prod_{k=1}^{M}(1-c_k z^{-1})}{\prod_{k=1}^{N}(1-d_k z^{-1})}$$

ROC:

$$|z|>\max_k |d_k|$$

$$H(z)=\underbrace{\left[\sum_{r=0}^{(M-N)} B_r z^{-r}\right]}_{\text{only if } M\geq N}+\sum_{k=1}^{N}\frac{A_k}{1-d_k z^{-1}}$$

$$h[n]=\sum_{r=0}^{(M-N)} B_r \delta[n-r]+\sum_{k=1}^{N} A_k d_k^n u[n]$$

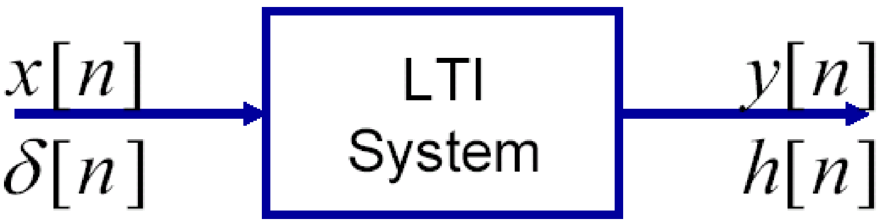
Frequency Response of a DE

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k]$$

$$H(e^{j\omega}) = H(z)\big|_{z=e^{j\omega}} = \frac{\left(\sum_{k=0}^{M} b_k e^{-j\omega k}\right)}{\left(\sum_{k=0}^{N} a_k e^{-j\omega k}\right)}$$

ROC must Contain the Unit circle

LTI System Characterization



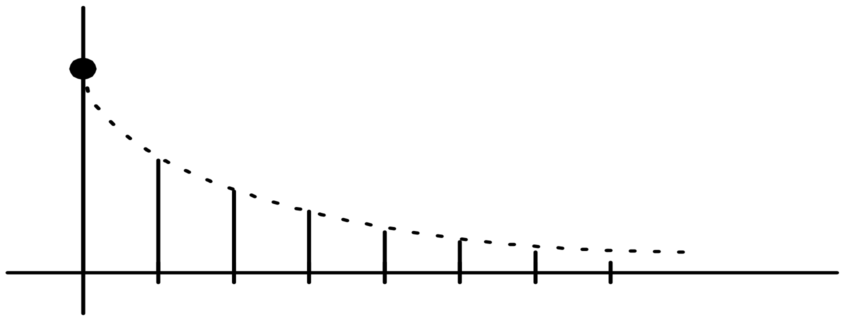
$x[n]$	$y[n]$
$\delta[n]$	$h[n]$
$e^{j\omega n}$	$H(e^{j\omega})e^{j\omega n}$
$X(e^{j\omega})$	$X(e^{j\omega})H(e^{j\omega})$
$X(z)$	$X(z)H(z)$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\left(\sum_{k=0}^{M} b_k z^{-k}\right)}{\left(\sum_{k=0}^{N} a_k z^{-k}\right)}$$

$$H(e^{j\omega}) = H(z)\big|_{z=e^{j\omega}}$$

Stability, Causality- illustration

(1) $x[n] = (\frac{1}{2})^n u[n]$



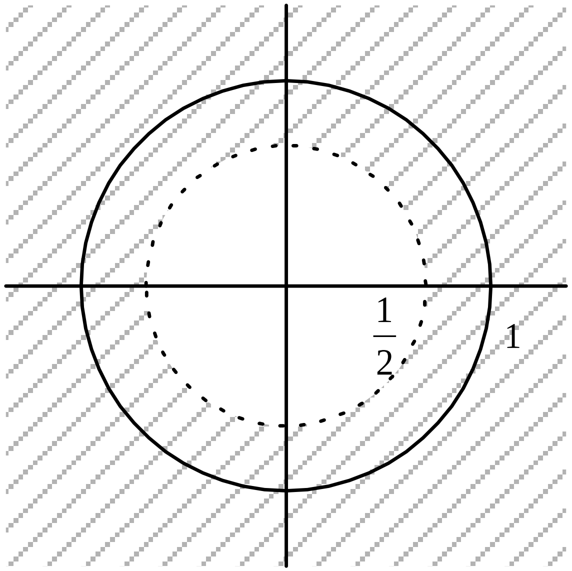
① Causal

②

. Stable

$$X(z) = \sum_{n=0}^{\infty} (\frac{1}{2})^n z^{-n} = \frac{1}{1 - \frac{1}{2} z^{-1}}$$

$$\text{RoC}: |z| > \frac{1}{2}$$

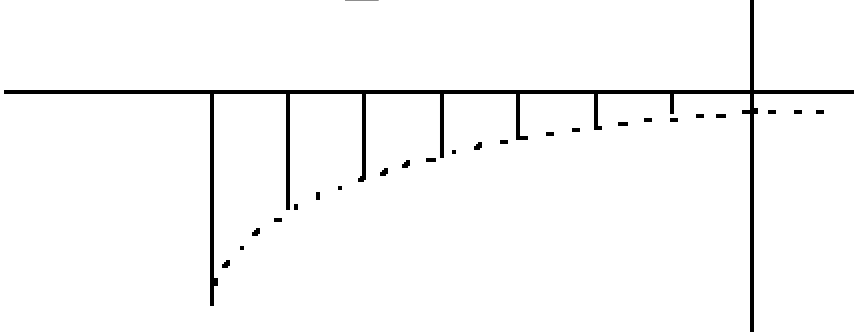


① Outward

② UC$\subset$ RoC

Stability, Causality- illustration..(cont)

(2) $x[n] = -(\frac{1}{2})^n u[-n-1]$



① Anti Causal

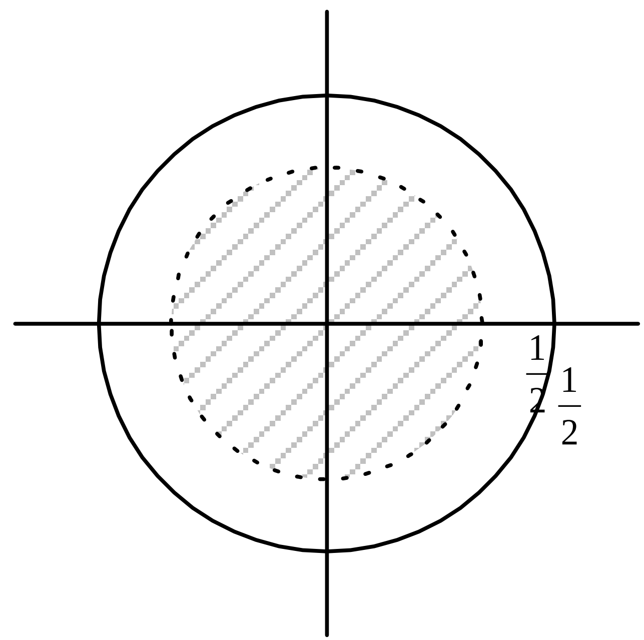
② Unstable

$$X(z) = \sum_{n=-\infty}^{-1} -(\frac{1}{2})^n z^{-n} = \frac{1}{1-\frac{1}{2}z^{-1}}$$

$$\text{ROC} : |z| < \frac{1}{2}$$

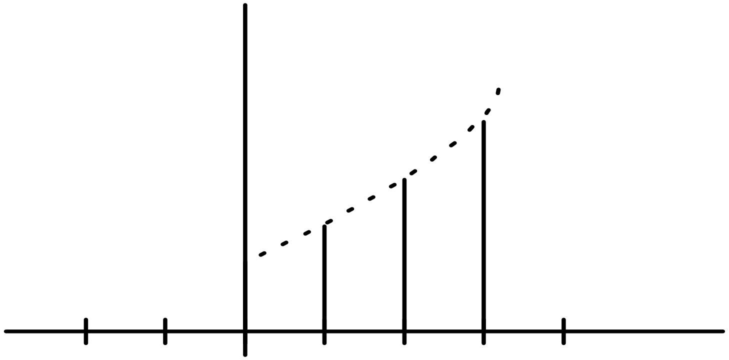
① Inward

② UC $\not\subset$ RoC



Stability, Causality- illustration..(cont)

(3) $x[n] = (2)^n u[n]$

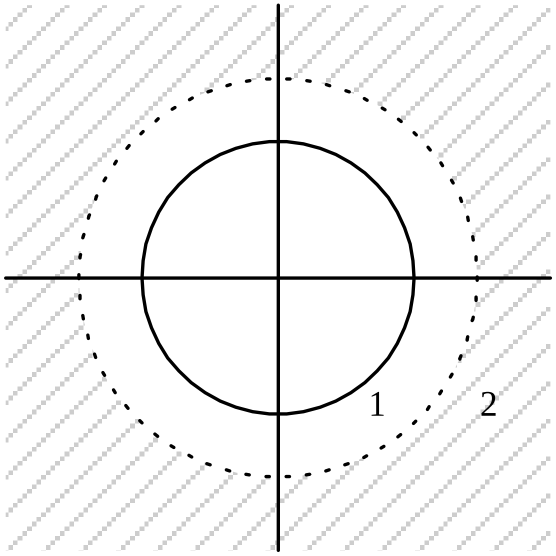


① Causal

② Unstable

$$X(z) = \sum_{n=0}^{\infty} 2^n z^{-n} = \frac{1}{1-2z^{-1}}$$

- RoC : $|z| > 2$

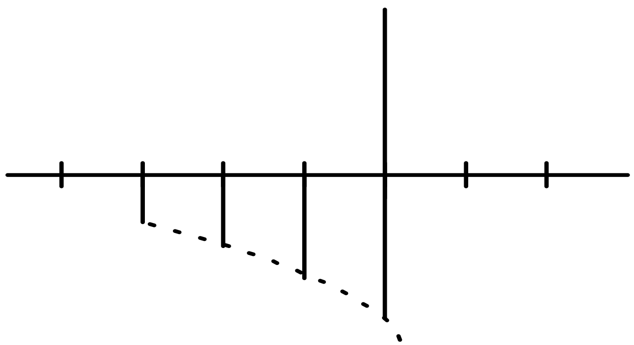


① Outward

② UC $\not\subset$ RoC

Stability, Causality- illustration..(cont)

(4) $x[n] = -2^n u[-n-1]$



$$X(z) = \sum_{n=-\infty}^{-1} -2^n z^{-n} = \frac{1}{1-2z^{-1}}$$

$\text{RoC} : |z| < 2$

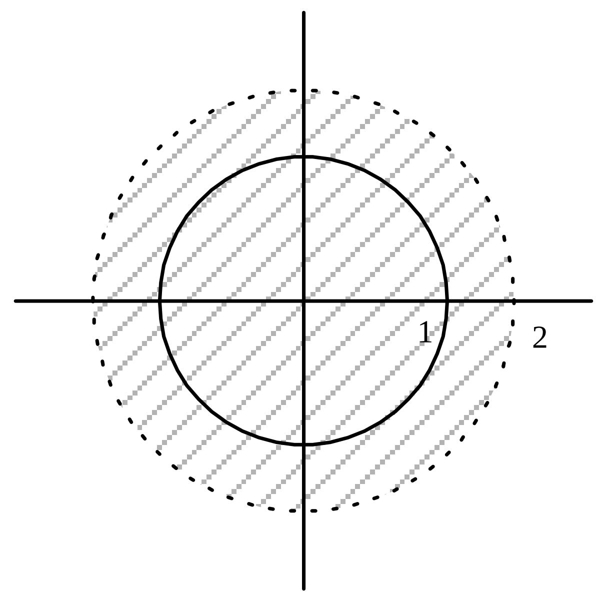
① Inward

② UC $\subset$ RoC

What do you find?

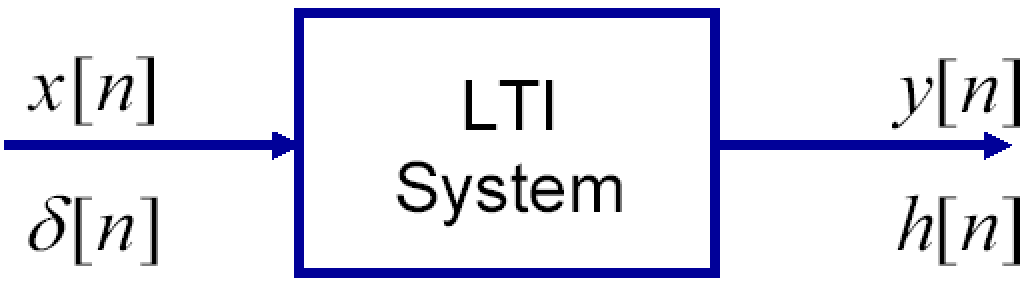
① Anti Causal

② Stable



Convolution Theorem

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] \Leftrightarrow Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega})$$



$e^{j\omega n}$ $\qquad$ $H(e^{j\omega})e^{j\omega n}$

$X(e^{j\omega})$ $\qquad$ $Y(e^{j\omega})$

$$Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega})$$

$$\left|Y(e^{j\omega})\right| = \left|H(e^{j\omega})\right|\left|X(e^{j\omega})\right|$$

$$\angle Y(e^{j\omega}) = \angle H(e^{j\omega}) + \angle X(e^{j\omega})$$

Rational system function

If $y[n]$ and $x[n]$ are related by

$$\sum_{k=0}^{M} a_k y[n-k] = \sum_{k=0}^{N} b_k x[n-k],$$

we have

$$H(z) = \frac{\sum_{k=0}^{N} b_k z^{-k}}{\sum_{k=0}^{M} a_k z^{-k}}$$

- stability/causality: depend on ROC–position of poles
- Causal & stable $\Leftrightarrow$ all poles in unit circle
- Inverse system: $H_i(z)H(z) = 1 \Leftrightarrow h[n] * h_i[n] = \delta[n]$. (ROCs must overlap)
- Causal & stable with Causal & stable inverse $\Leftrightarrow$ all poles and zeros in unit circle

Inverse System

$$H(z)H_i(z) = 1 \qquad h[n] * h_i[n] = \delta[n]$$

$$H_i(z) = \frac{1}{H(z)} = \frac{b_0}{a_0} \cdot \frac{\prod_{k=1}^{M}(1 - d_k z^{-1})}{\prod_{k=1}^{N}(1 - c_k z^{-1})}$$

Inverse system, stable if all poles and zeros, inside the unit circle

→ **<u>minimum-phase system</u>**

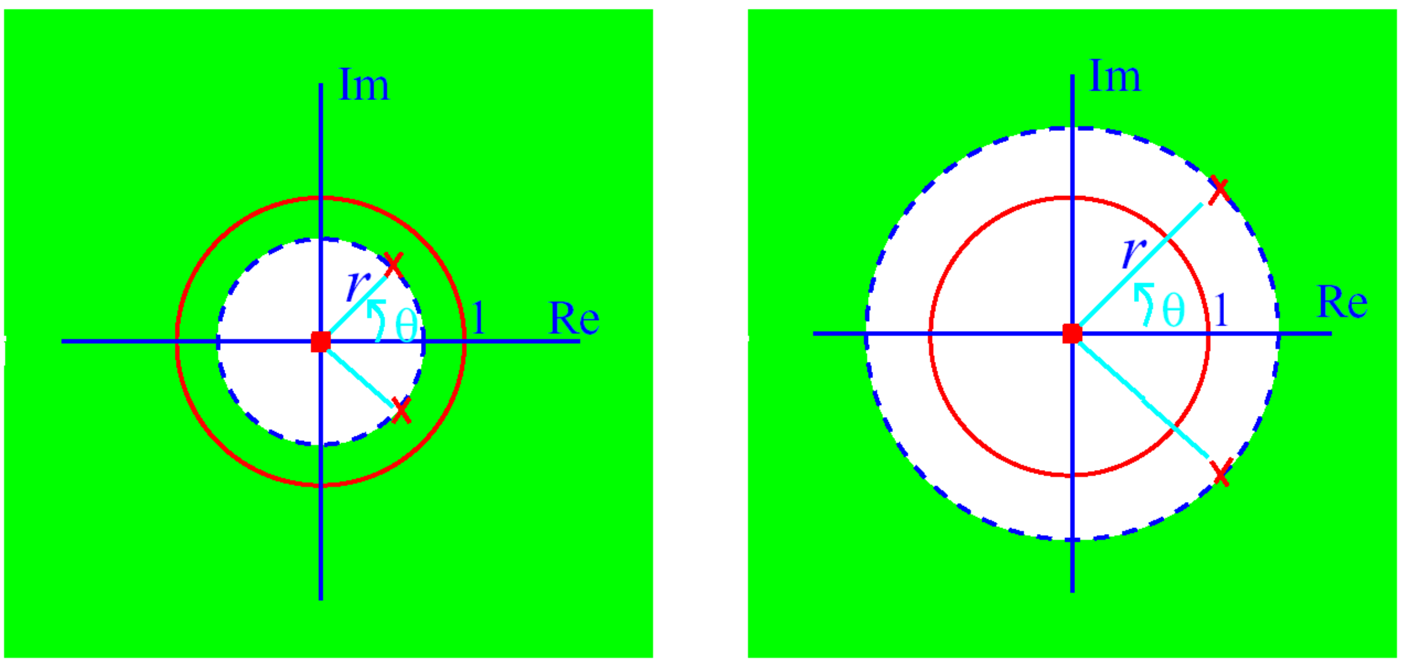
$$\text{(e.q)} \quad H(z) = \frac{1 - 0.5z^{-1}}{1 - 0.9z^{-1}}, \quad h(n) = (0.9)^n u(n) - 0.5(0.9)^{n-1} u(n-1)$$

$$H_i(z) = \frac{1 - 0.9z^{-1}}{1 - 0.5z^{-1}}, \quad h_i(n) = (0.5)^n u(n) - 0.9(0.5)^{n-1} u(n-1)$$

Stability

Example:

$$H(z) = \frac{1}{1 - [2r\cos\theta]z^{-1} + r^2 z^{-2}}$$



stable system $(r<1)$ **unstable system** $(r>1)$

Group Delay

Phase distortion is usually measured by a quantity called Group Delay

Definition:

-The negative of the derivation of the phase response with respect to the frequency:

$$\tau(\omega) = -\frac{d}{d\omega}\left[\angle H\left(e^{j\omega}\right)\right]$$

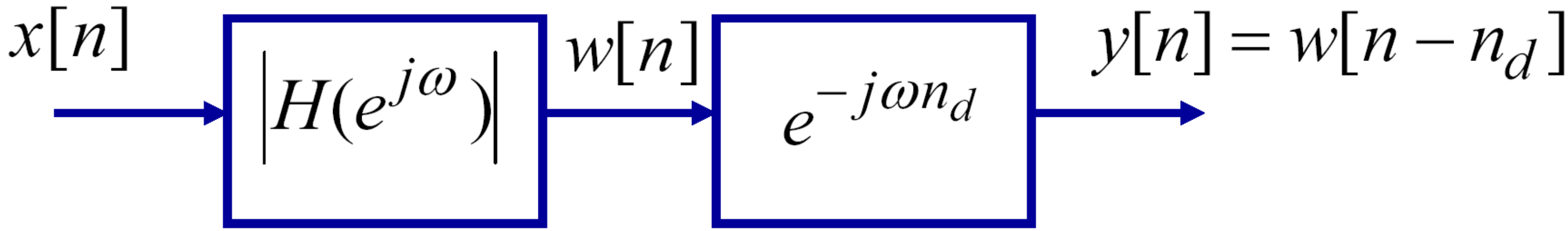
->The deviation of the Group Delay away from constant indicates the degree of non-linearity of the phase, thus giving a measure of phase distortion. We want the Group Delay to be as constant as possible.

Group Delay

$$\tau(\omega) = \mathrm{grd} H(e^{j\omega}) = -\frac{d}{d\omega} \arg H(e^{j\omega})$$

$$(e.q) \arg H(e^{j\omega}) = -\alpha\omega \quad \Rightarrow \quad \tau(\omega) = \alpha$$

Linear Phase Means Delay



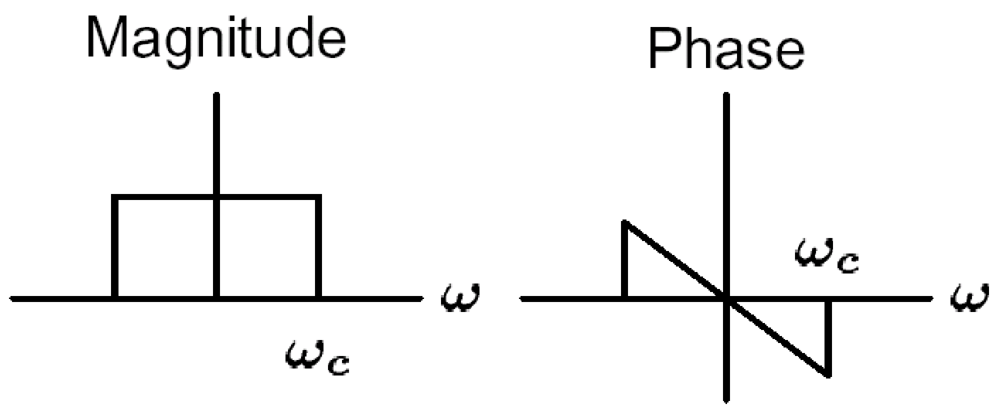
- Since we can consider the effects of magnitude and phase separately, it follows that linear phase of the form $\angle H(e^{j\omega}) = -\omega n_d$ implies delay of n_d samples.

- Thus an ideal lowpass filter with delay has

$$h_{\mathrm{lp}}[n] = \frac{\sin \omega_c (n - n_d)}{\pi (n - n_d)} \Leftrightarrow H(e^{j\omega}) = \begin{cases} e^{-j\omega n_d}, & |\omega| < \omega_c \\ 0, & \omega_c < |\omega| < \pi \end{cases}$$

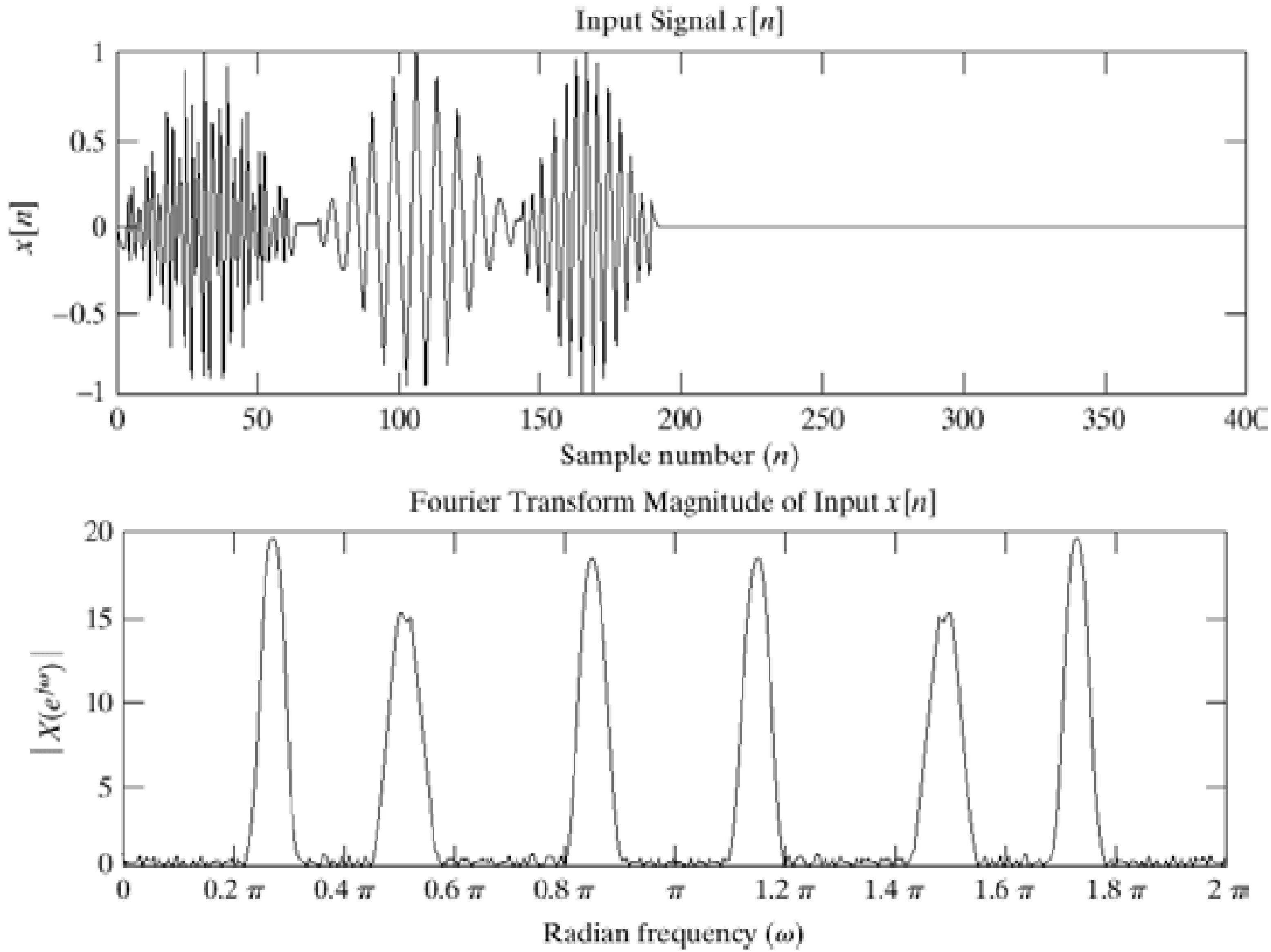
Example: Ideal lowpass filter frequency response

$$H(e^{j\omega}) = \begin{cases} e^{-j\omega n_d}, & |\omega| < \omega_c, \\ 0, & \text{otherwise.} \end{cases}$$

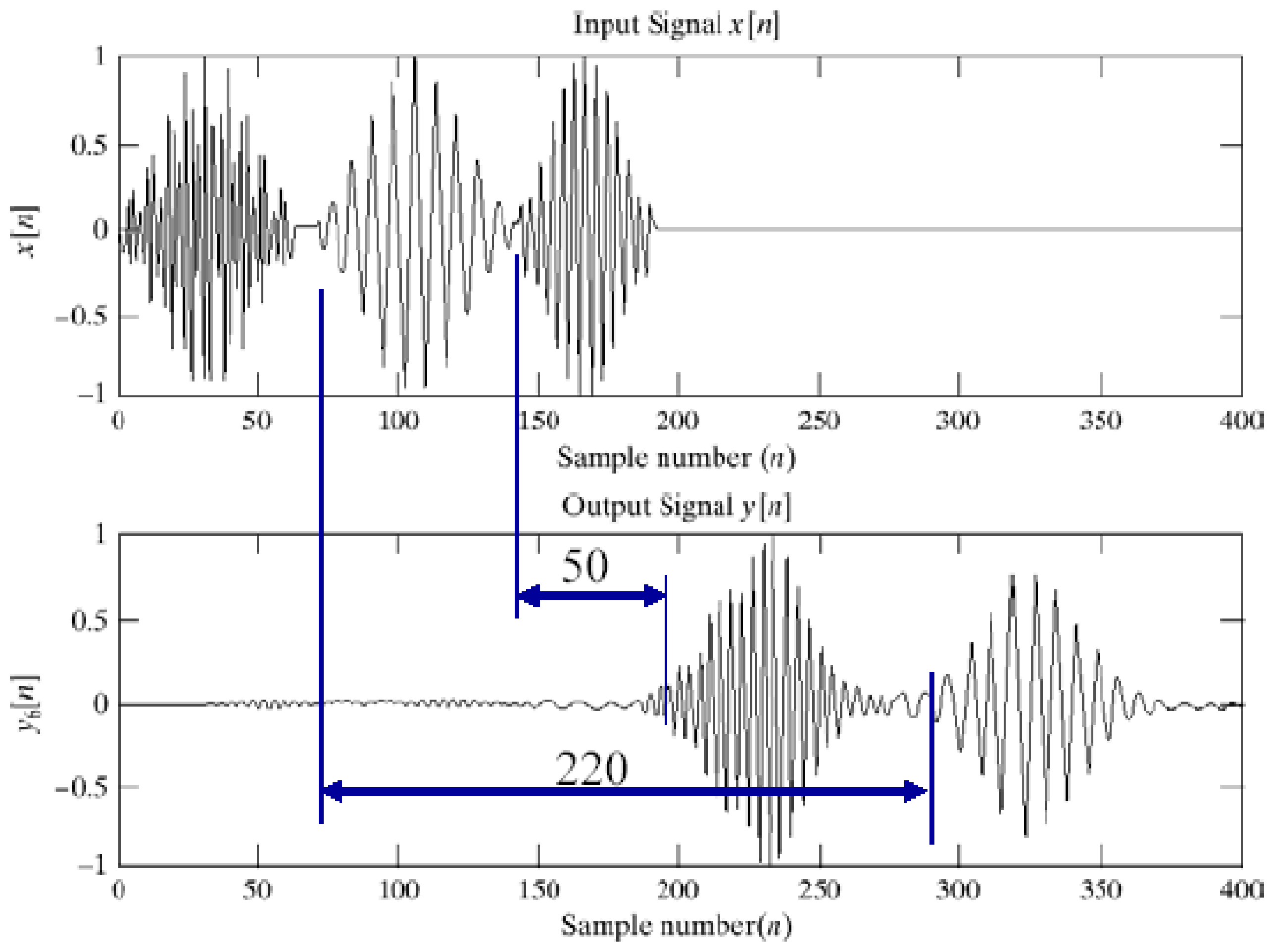


$$h[n] = \frac{\sin \omega_c (n - n_d)}{\pi (n - n_d)}$$

Input Signal



Output Signal



Ideal Lowpass with Delay

- Frequency response:

$$H(e^{j\omega}) = \begin{cases} e^{-j\omega\alpha} & |\omega| < \omega_c \\ 0 & \omega_c < |\omega| \le \pi \end{cases}$$

- Magnitude and phase:

$$\left|H(e^{j\omega})\right| = \begin{cases} 1 & |\omega| < \omega_c \\ 0 & \omega_c < |\omega| \le \pi \end{cases} \qquad \angle H(e^{j\omega}) = -\omega\alpha$$

- Impulse response:

$$h[n] = \frac{\sin \omega_c (n-\alpha)}{\pi(n-\alpha)} \qquad -\infty < n < \infty$$

Example

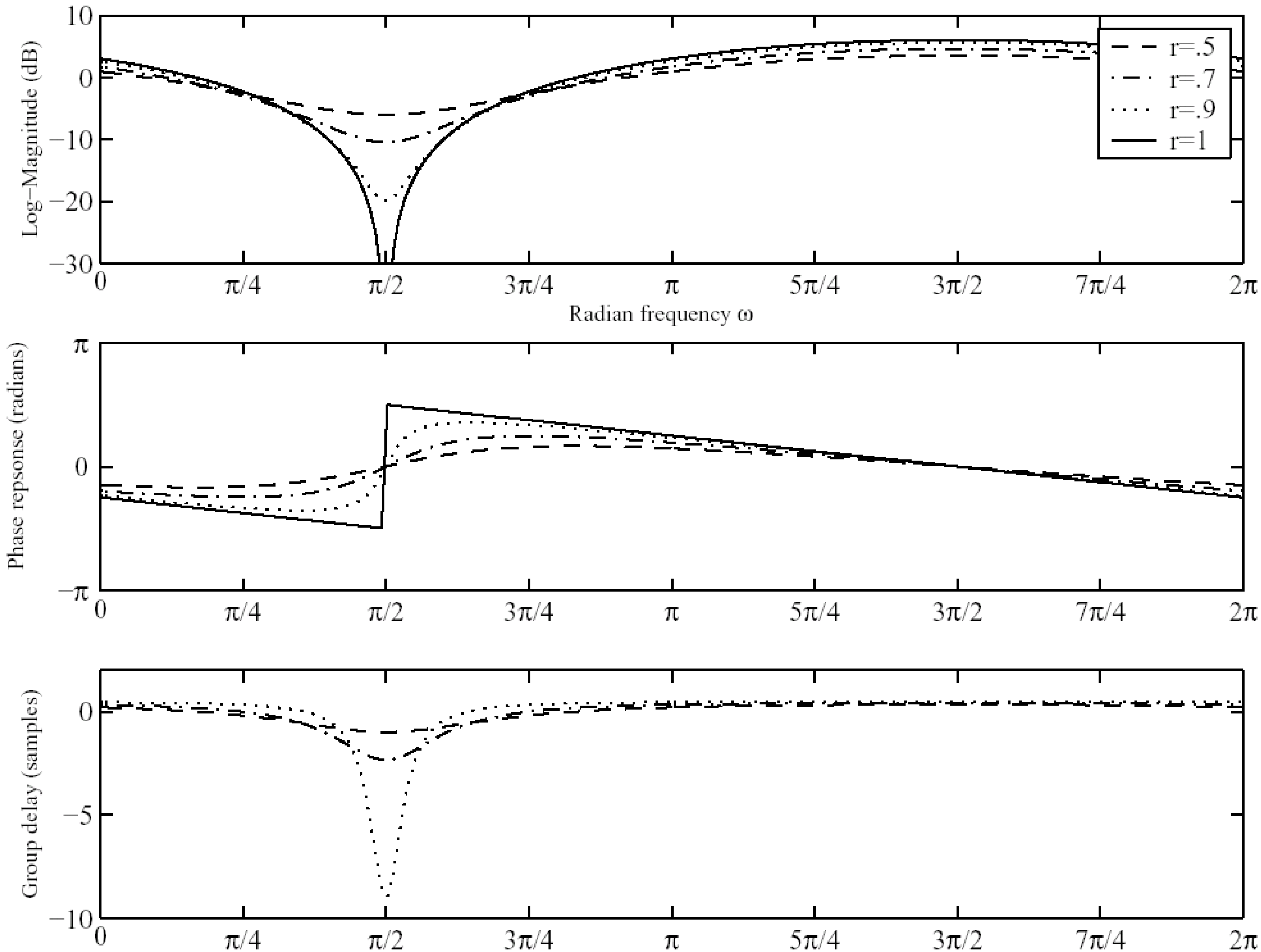
$$H(z) = 1 - re^{j\theta} z^{-1}$$

$$H(e^{j\omega}) = 1 - re^{j\theta} e^{-j\omega}$$

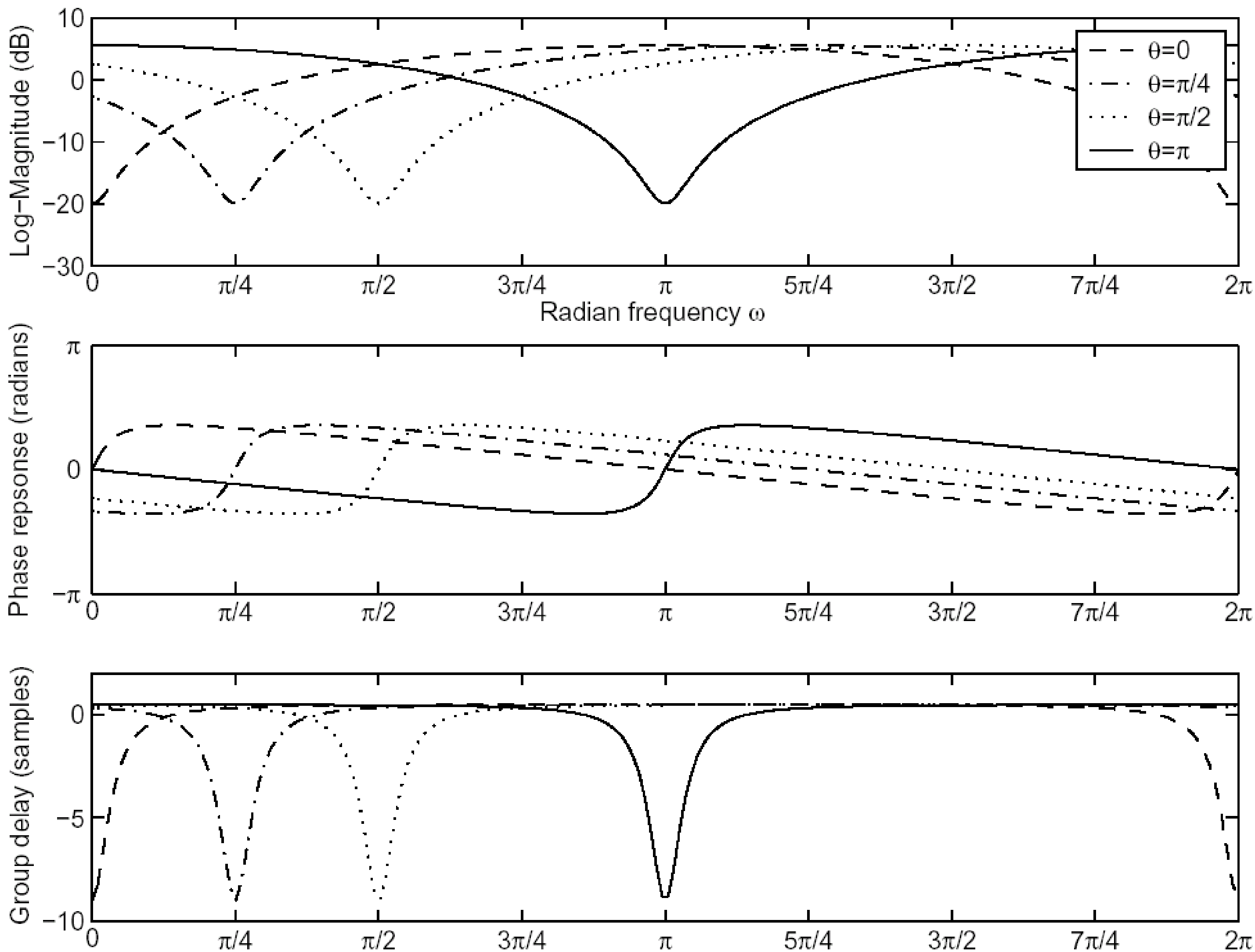
$$\begin{cases} |H(e^{j\omega})| = 10\log[1 + r^2 - 2r\cos(w-\theta)] \\ \angle H(e^{j\omega}) = \arctan \dfrac{r\sin(w-\theta)}{1 - r\cos(w-\theta)}] \\ \text{grd}\, H(e^{j\omega}) = \dfrac{r^2 - r\cos(w-\theta)}{1 + r^2 - 2r\cos(w-\theta)} \end{cases}$$

Frequency response of one zero system

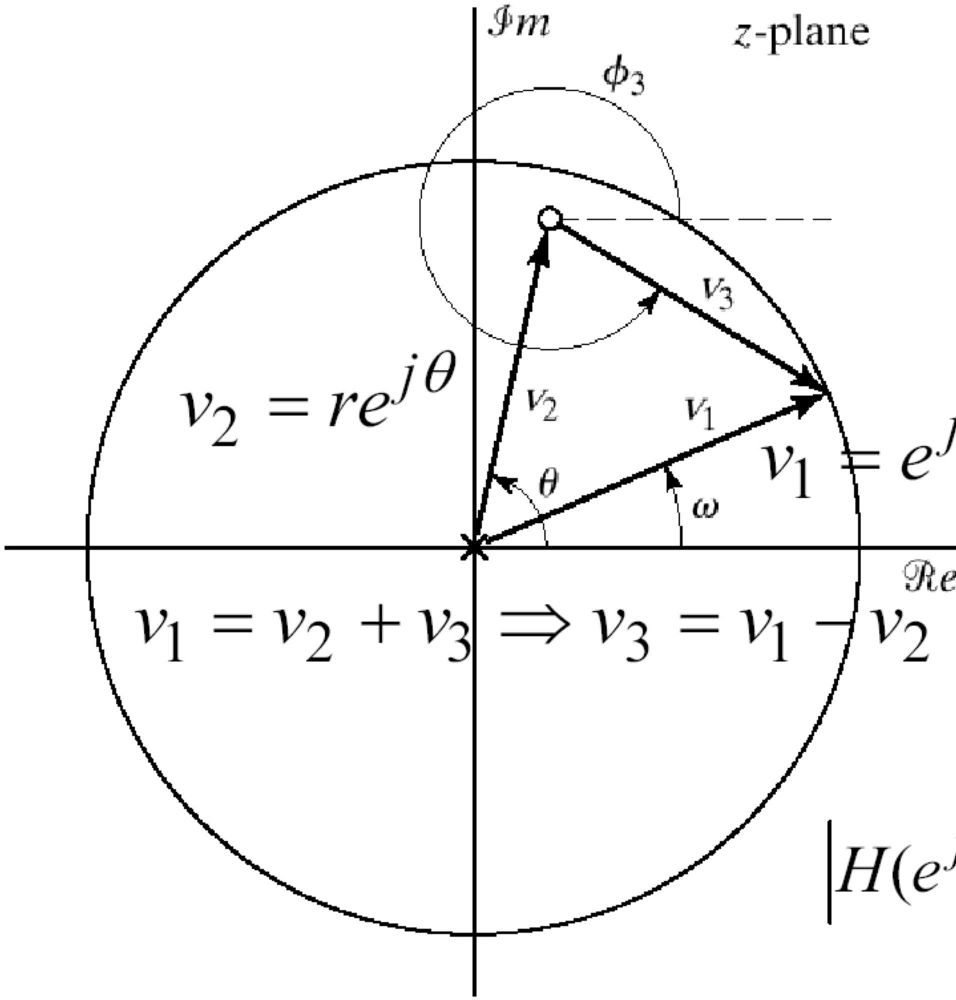
zero at $re^{j\theta}$



Frequency response of one zero system



Pole Zero Plot



$$v_2 = re^{j\theta}$$

$$v_1 = e^{j\omega}$$

$$v_1 = v_2 + v_3 \Rightarrow v_3 = v_1 - v_2$$

$$H(z) = (1 - re^{j\theta} z^{-1})$$

$$= \frac{z - re^{j\theta}}{z}$$

$$H(e^{j\omega}) = \frac{e^{j\omega} - re^{j\theta}}{e^{j\omega}}$$

$$= \frac{v_3}{v_1}$$

$$\left|H(e^{j\omega})\right| = \frac{\left|e^{j\omega} - re^{j\theta}\right|}{\left|e^{j\omega}\right|} = \frac{|v_3|}{|v_1|} = |v_3|$$